

1. (7 points)

An insurance company historically never attempted to recover salvage and subrogation (“S&S”) on claims. On January 1, 2018, the insurer enters into a one-year agreement with an S&S recovery vendor, requiring the vendor to pursue all S&S opportunities for accident year 2018.

The insurance company’s IT department generated the following loss development triangles as of December 31, 2018 (assume no development after 36 months):

Incremental Paid Loss Gross of S&S (\$000s) as of (months)				Incremental Paid Loss Net of S&S (\$000s) as of (months)			
Accident Year	12	24	36	Accident Year	12	24	36
2016	16,500	6,000	2,500	2016	16,500	6,000	2,500
2017	17,000	5,000		2017	17,000	5,000	
2018	14,000			2018	11,000		

Prior to 2018, the Actuarial department estimated unpaid losses using the following Bayesian model:

- C_{ij} represents the incremental losses for accident year i as of development year j which follow an over-dispersed Poisson (“ODP”) distribution with mean $x_i y_j$ and variance $\phi x_i y_j$.
- x_i represents the expected ultimate losses for accident year i .
- y_j represents the proportion of ultimate losses that emerge in development year j .
- The prior distribution for x_i is gamma with mean $\frac{\alpha_i}{\beta_i}$ and variance $\frac{\alpha_i}{\beta_i^2}$.
- ϕ represents the dispersion parameter for the ODP distribution, which is 93.
- λ_j represents the incremental chain ladder loss development factor for development year j .
- D_{ij} represents the cumulative losses for accident year i as of development year j .
- M_i represents the value for ultimate losses for accident year i that is obtained using expert knowledge about the losses.
- The mean of C_{ij} for this Bayesian model is:

$$Z_{ij} (\lambda_j - 1) D_{i,j-1} + (1 - Z_{ij}) (\lambda_j - 1) M_i \frac{1}{\lambda_j \lambda_{j+1} \dots \lambda_n}, \text{ where } Z_{ij} = \frac{\sum_{k=1}^{j-1} y_k}{\beta_i \phi + \sum_{k=1}^{j-1} y_k}.$$

The Actuarial department believes that the S&S recovery vendor agreement has resulted in a slowdown of the gross loss payment pattern for calendar year 2018 and will continue in calendar years 2019 and 2020.

To estimate the total unpaid losses gross of S&S for accident year 2018, the Actuarial department is considering the following prior gamma distributions for the Bayesian model above:

Parameter	Option 1	Option 2	Option 3
α	50	200	1,250
β	0.002	0.008	0.100

To estimate the total S&S recoveries for accident year 2018, the Actuarial department analyzed S&S data from competitors of comparable size to find:

- The expected S&S recoveries per accident year were \$5M.
- The standard deviation of the S&S recoveries was \$1M.
- The expected percent of the S&S recoveries received through 12 months was 75%.
- The standard deviation of the percent of S&S recoveries received through 12 months was 10%.

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1. (continued)

a. (1 point)

Select the prior gamma distribution most appropriate to address the concern that the vendor agreement has slowed the gross loss payment pattern. Justify the selection.

b. (2 points)

Calculate the incremental losses gross of S&S for accident year 2018 expected to emerge between 12 and 36 months using the Bayesian model with the most appropriate prior distribution from part a. above.

c. (0.5 point)

Describe model risk with respect to the Bayesian model for incremental losses gross of S&S.

d. (1.5 points)

Calculate the total S&S recoveries for accident year 2018 using Bayesian credibility.

e. (0.5 point)

Describe estimation risk with respect to the Bayesian model for S&S recoveries.

f. (0.5 point)

Calculate the total unpaid losses net of S&S for accident year 2018.

g. (1 point)

Describe two types of operational risk introduced by the vendor agreement and recommend a unique key risk indicator ("KRI") to monitor each risk.

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QUESTION 1	
TOTAL POINT VALUE: 7.0	LEARNING OBJECTIVE(S): A1, A8, A10, C6, C7
SAMPLE ANSWERS	
Part a: 1 point	
<u>Sample 1</u> Mean (α/β): Option 1: 25,000 Option 2: 25,000 Option 3: 12,500 Variance ($\frac{\alpha}{\beta^2}$) Option 1: 12,500,000 Option 2: 3,125,000 Option 3: 125,000 Select Option 2. Mean value is more reasonable. Variance is smaller which gives less credibility to LDF, and more credibility to prior estimate. <u>Sample 2</u> We want to select Option 2. Option 3 results in an expected loss that is half of the other options at 12,500. Based on historical data, an ultimate loss estimate of 25,000 is more reasonable. Option 2 has higher beta vs Option 1. Thus results in a smaller variance, stronger prior. We want to rely more on our expert opinion and less on historical data. So Option 2 provides the most weight on our expert opinion. <u>Sample 3</u> Expected values: Option 1: $50/0.002 = 25,000$ Option 2: 25,000 Option 3: 12,500 Since 2018 is already at 14M as gross loss, I selected Option 1 or 2. Assuming I am confident in my prior estimate, I choose the one with higher β. Picking Option 2.	
Part b: 2 points	
<u>Sample 1</u> Calculated LDFs: $12-24 \text{ LDF} = (22500 + 22000)/(16500 + 17000) = 1.3284$ $24-36 \text{ LDF} = 25000/22500 = 1.111$ $\text{LDF}_{12} = 1.111 * 1.328 = 1.475$, $\% \text{paid}_{12} = 1/1.475 = 0.678$ $\text{LDF}_{24} = 1.111$, $\% \text{paid}_{12} = 1/1.111 = 0.900$	

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$$\text{Calculate } Z_{12} = 0.678 / (0.008 * 93 + 0.678) = 0.477$$

$$Z_{24} = 0.900 / (0.008 * 93 + 0.900) = 0.547$$

$$E[\text{incremental loss at 24 mo}] = 0.477 * 14000 * (1.328 - 1) + (1 - 0.477) * 25000 * (0.9 - 0.68) = 5093$$

$$E[\text{incremental loss at 36 mo}] = 0.547 * (1.111 - 1) * (14000 + 5093) + (1 - 0.547) * 25000 * (1 - 0.9) = 2291.77$$

$$\text{Total expected emergence for AY 2018} = 5093 + 2291.77 = 7384.77 \text{ (in 000s)}$$

Sample 2

$$E(\text{Ult}) = 25000$$

$$\text{LDF } 12-24 = (16500 + 6000 + 17000 + 5000) / (16500 + 17000) = 1.328$$

$$24-36 = 1.111$$

$$12-\text{ult} = 1.475$$

$$\% \text{ rept at 12} = 1 / (1.328 * 1.111) = 68\%$$

$$z = (\% \text{ rept}) / (\text{beta} * \phi + \% \text{ rept}) = 0.68 / (0.008 * 93 + 0.68) = 0.478$$

$$0.478 * (1.475 - 1) * (14000) + (1 - 0.478) * (1.475 - 1) * (25000) * (0.68) = 7394$$

Sample 3

$$\text{Gamma Mean} = 25000$$

$$\text{LDFs: Age 12} = 1.328, \text{ Age 24} = 1.111$$

$$\text{CLDFs: Age 12} = 1.475, \text{ Age 24} = 1.111$$

$$\% \text{ Paid: Age 12} = 0.678, \text{ Age 24} = 0.9$$

$$\% \text{ Emerged: Age 12-24} = 0.222, \text{ Age 24-36} = 0.1$$

$$Z_{12} = 0.678 / (0.008 * 93 + 0.678) = 0.477$$

$$\text{CL Ultimate} = 20649$$

$$\text{CL}_{12-24} = 4584, \text{ CL}_{24-36} = 20649 * 0.1 = 2065$$

$$\text{BF}_{12-24} = 5500, \text{ BF}_{24-36} = 25000 * 0.1 = 2500$$

$$\text{Wtd}_{12-24} = 5063, \text{ Wtd}_{24-36} = 2065 * (0.477) + 2500 * (1 - 0.477) = 2292.5$$

Part c: 0.5 point

Sample 1

1. The prior distribution of Gamma might not be true
2. The incremental loss might not follow an ODP distribution
3. Bayesian model won't perfectly reflect the process. Model risk is the uncertainty that the model won't perfectly reflect the insurance process

Sample 2

Model risk is risk of not specifying the correct model. In Bayesian model, we weigh our faith in the specified model via the β parameter, which controls how much weight is given to the model (chain ladder) versus our a priori estimate. Higher β reduces the credibility measure, thus giving more weight to our a priori estimate.

Sample 3

The competitor experience may not apply to the insurer. If it is the case, the prior distribution selected may not be accurate. And hence the model can't incorporate the expert opinion very well.

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Sample 4

Model risk is the risk of selecting the incorrect loss distribution for the a priori expectation or that we are not using the correct credibility procedure.

Sample 5

The Bayesian model is putting low credibility on actual experience and high credibility on the gamma distribution estimate. This is risky to put a lot of credibility on the assumption of a slow down in a green year with no historical data to back it up.

Part d: 1.5 point

Sample 1

$$\text{VHM} = \text{Var} [E(X/Y)] = d^2 \cdot \sigma^2 = (0.75)^2 \times 1^2 = 0.563$$

$$\text{EVPV} = E [\text{Var} (X/Y)] = \sigma_d^2 (\sigma_Y^2 + E(Y)^2) = (0.1)^2 \times (1^2 + 5^2) = 0.26$$

$$Z = 0.563 / (0.563 + 0.26) = 68.4\%$$

$$\text{Ult}_{SS} = 68.4\% \times [(14 - 11) / 0.75] + (1 - 68.4\%) \times 5 = 4.31 \text{ million}$$

Sample 2

$$E(Y) = 5,000,000 \quad \sigma(Y) = 1,000,000 \quad E(X/Y) = 0.75 \quad \sigma(X/Y) = 0.1$$

$$\text{VHM} = [0.75 (1,000,000)]^2 = 5.625 \times 10^{11}$$

$$\text{EVPV} = .12 (5M^2 + 1M^2) = 2.6 \times 10^4$$

$$Z = \text{VHM} / (\text{VHM} + \text{EVPV}) = .684$$

$$\text{S\&S to date } (\$000) = 14,000 - 11,000 = 3,000$$

$$\text{S\&S Recov for AY 2018 } (\$000) = 3,000 / .75 \times (.684) + (1 - .684)(5,000) = \$4,316$$

Part e: 0.5 point

Sample 1

- Estimation risk is the risk that the forms and parameters chosen don't reflect the "true" form and parameters, due to estimating from a sample of the data.
- For S&S, the department took data from competitors of comparable size. However, there could be a difference in their mix of business, their credibility of data, and assumptions/definitions that can cause our estimates from it to misrepresent the data.

Sample 2

As data obtained from competitors, the $E(\text{S\&S})$ and σ might be wrong for our company (different make-ups, lines, practice, etc.)

Sample 3

Estimation risk is risk we have not correctly estimated a parameter for the loss emergence, and specifically for this, there is a risk that our estimate of $E[X/Y]$ (% S&S emerged @12mo.) is inaccurate, especially since we don't have our own past data to go on, only competitor data which might not reflect our business.

Sample 4

Estimation risk – the risk that the selected parameters of the model are not the "true" parameters of the distribution, would result from the appropriateness of the selected competitors for comparison regarding the similarity in lines of business, states written and vendor selection to the company.

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Part f: 0.5 point

Sample 1

Future Expected Recoveries = $4.316 - 3M = 1.316M$

Total Unpaid Net of S&S = $7.396M - 1.316M$
= $6.08M$

Sample 2

AY 2018 Ultimate Loss Gross S&S = $14M + 7.395M = 21.395M$

AY 2018 Total S&S = $4.316M$

AY 2018 Ultimate Loss Net S&S = $21.395M - 4.316M = 17.079M$

Paid Loss Net S&S = $11M$

Unpaid Net S&S = $6.079M$

Part g: 1 point

Any two of the following samples:

Clients & Business Practices – not meeting professional obligations to clients; could be trying to over recover and receiving complaints to the DOI. (KRI – monitor number of complaints)

External Fraud – S&S company may put falsified information into the claims and return less money to the company (KRI – external auditor's reports)

Execution & Process Change Management – employees have never done S&S before; this could lead to people getting upset over the added work or over glitches that may occur during implementation (KRI – monitor employee turnover ratio)

Internal Fraud – since there will be an increased number of transactions with another company, there may be more opportunities for accounting people to slip in fraudulent transactions to steal money from the insurer (KRI – keep track of the number of ledger entries for transactions with the vendor and look for any unexpected increases)

Recovery Risk – comes from uncertainty of recovery which could be a significant amount of total gross claim (KRI – recovery amount / total gross claim amount)

System Failure – integrating our systems within the new S&S processes could cause problems, perhaps from tying our reporting systems & the vendors payments, and tracking systems; this could possibly cause processing center down time (KRI – processing center down time)

EXAMINER'S REPORT

Candidates were expected to evaluate models across multiple areas of risk, and select the most appropriate model based on the relationship between model parameters, desired credibility, and changes in the expected claims environment. Candidates were also expected to apply prior distributions and mean/variance assumptions to estimate reserves after appropriately blending this information with the indications implied by historical claims development.

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Part a

Candidates were expected to demonstrate knowledge with identifying data issues and related model adjustments for reserving models, as well as testing assumptions underlying reserving models.

Common mistakes included:

- Selected Option 3 with the lowest variance (the highest β) without considering the reasonability of mean under Option 3
- Correctly selected Option 2 with lower variance, however, incorrectly linked lower variance (higher β) to more weight to Chain Ladder (weaker prior), or incorrectly linked lower variance to reflect slow down payment pattern
- Incorrectly selected Option 1 with larger variance (lower β) to put less weight on prior estimate
- Selected Option 2 as the variance is in the middle of three options
- Calculated standard deviation from three options, and selected Option 2 as it is the closest to competitor's standard deviation of 1M
- Calculated variance using Accident Year 2016 and 2017 ultimate losses, and selected Option 2 as it has the closest variance
- Mean or variance were calculated incorrectly

Part b

Candidates were expected to combine information from the loss triangle and prior gamma distribution to calculate future incremental payments based on the provided credibility formula.

Common mistakes include

- Using the Net of S&S instead of the Gross of S&S triangle
- Estimating %pd as the reciprocal of age-to-age LDFs instead of age-to-ultimate LDFs
- Only estimating incremental paid losses for 12-24 months and not 12-36 months
- When using different credibility weights for the 12-24 and 24-36 periods, using the wrong estimate for $D_{2018,24}$ in the credibility formula, most commonly by using the CL method indicated incremental losses instead of applying the CL development factors to $14k + \text{the credibility wtd mean of } C_{2018,24}$
- Using incremental %pd amounts instead of cumulative %pd amounts in the calculation of z factors
- Using one Gamma prior distribution to calculate M_i and a different Gamma prior distribution to estimate Z
- Applying LDFs to M_i without adjusting M_i for the %pd implied by the LDFs
- When applying the credibility formula to ultimate loss estimates instead of reserve estimates, candidates often applied the complement of credibility to the prior Gamma mean instead of the BF method (i.e. $14k + \text{Gamma Apriori} * \% \text{unpaid}$)

Part c

Candidates were expected to define model risk and relate it to the context of this problem's Bayesian model for losses gross of S&S. However, full credit was also given to candidates who explained how model risk related to this problem without defining it, as the question did not explicitly ask for a definition.

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Common mistakes include • Failing to link model risk back to the question, and only defining it • Only discussing S&S model risk • Discussing estimation risk. Although estimation risk is a subset of model risk, estimation risk is asked directly later in the problem.
Part d Candidates were expected to understand and apply a Bayesian credibility method as presented in the Brosius paper to the specific situation described in this question and provide an ultimate S&S estimate using that approach. Common mistakes included: • Errors in calculating the VHM and EVPV values, either from applying the wrong formulas, using the incorrect inputs or having calculation errors. • Errors in credibility weighting a developed loss amount and an expected loss amount. These included not properly developing the \$3M of S&S paid to date to an ultimate value or not incorporating the appropriate expected ultimate amount into the formula presented in the syllabus reading.
Part e Candidates were expected to understand the estimation risk, and why the insurer would have estimation risk from using only competitor's data. Common mistakes included: • Only provided definition of estimation risk • Failed to comment on insurer using competitor's data to estimate the S&S parameters
Part f Candidates were expected to recognize that the total unpaid losses gross of S&S was calculated in part b. and the ultimate S&S recoveries was calculated in part d., and then to combine those responses to come up with the total unpaid losses net of S&S. Common mistakes included: • Not subtracting the paid-to-date losses • Calculating the ultimate losses net of S&S instead of the unpaid losses net of S&S • Using the Bayesian model for losses gross of S&S to estimate the losses net of S&S
Part g Candidates were expected to define two operational risks and discuss how the risk is related to the agreement with the S&S vendor. Also, candidates were expected to identify quantifiable KRIs that related to the operational risk described. Common mistakes include: • Identifying an operational risk, such as 'External Fraud', but failing to describe the risk • Providing insufficient detail for a KRI, such as stating 'monitor S&S' • Identifying the following as operational risks. <u>Expense risk</u> – risk that expenses increase as a result of vendor agreement

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- o Expenses are expected to increase with the agreement. An operational risk would be if the expenses increased more than expected

Reserving process risk – slowing down the claim payment patterns and affecting reserve projections

- o It was stated in the scenario that claim payment patterns are slowing. An operational risk would be if this slowdown upsets insureds and impacts customer retention